MASTER'S OF COMPUTER APPLICATION

LOVELY PROFESSIONAL UNIVERSITY

A

Project Report on

Women's Safety & Security System

"रक्षा कवच (Protection Shield)"

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Table of Contents

Chapter 1: Introduction

1.1 Introduction..... 3
1.2 Objectives..... 3
1.3 Needs of Women's Safety System..... 3
1.4 Methodology Development Model..... 3
1.5 Tools and Technique..... 4
1.6 External Interface..... 5

Figure: Schema Diagram..... 6

Chapter 2: Task and Activities Performed

2.1 Profile of Problems..... 7

2.2 Structure of the Project..... 7

2.3 Scope and Feasibility..... 9

2.4 System Design..... 9

2.5 Implementation..... 10

2.6 Test Generation..... 10

2.7 Screen Shots..... 12

2.8 Conclusion..... 22

Chapter 1: Introduction

1.1 Introduction

The Women Safety & Security System is an innovative web application designed to enhance the safety and security of women in society. By providing a platform for immediate assistance, the application facilitates quick communication with authorities and guardians, thereby promoting a sense of security among users.

1.2 Objective

The primary objective of this application is to empower women by providing them with a reliable tool to report incidents, seek help, and keep their guardians informed in emergencies. The system aims to create a safer environment for women through technology.

1.3 Needs of Women Safety System

In recent years, the need for women safety systems has grown exponentially due to the increase in incidents of harassment and violence against women. This application addresses the following needs:

- **Immediate Assistance:** Quick access to help during emergencies.
- **Communication:** Efficient communication between users and their guardians or law enforcement.
- **Awareness:** Raising awareness of safety measures and reporting mechanisms.

1.4 Methodology Development Model

The development of this application followed the Agile methodology, allowing for iterative development, constant feedback, and flexibility in adapting to changes. The key phases included:

- Plan
- Design
- Development
- Testing
- Deployment
- Review
- Launch

1.5 Tools and Technique

The application was built using a combination of modern technologies and tools:

Frontend Development:

- **HTML:** For structuring content.
- **CSS:** For styling and layout.
- **Bootstrap:** For responsive design, ensuring usability on various devices.

Backend Development:

- **PHP:** For server-side scripting to handle user interactions and data management.
- **MySQL:** For relational database management, storing user data, complaints, and notifications securely.

APIs:

- **Twilio API:** Used for sending SMS notifications to users' guardians in emergencies, enhancing the application's communication capabilities.

Version Control:

- **Git:** Employed for version control, allowing for efficient collaboration among team members and ensuring a history of code changes.

1.6 External Interfaces

This interface will be the actual interface through which the user communicates with the application and performs the desired tasks. The footer has one function, "About application & Developer." If the user wants to read the documentation and learn about the application, he/she can do so easily.

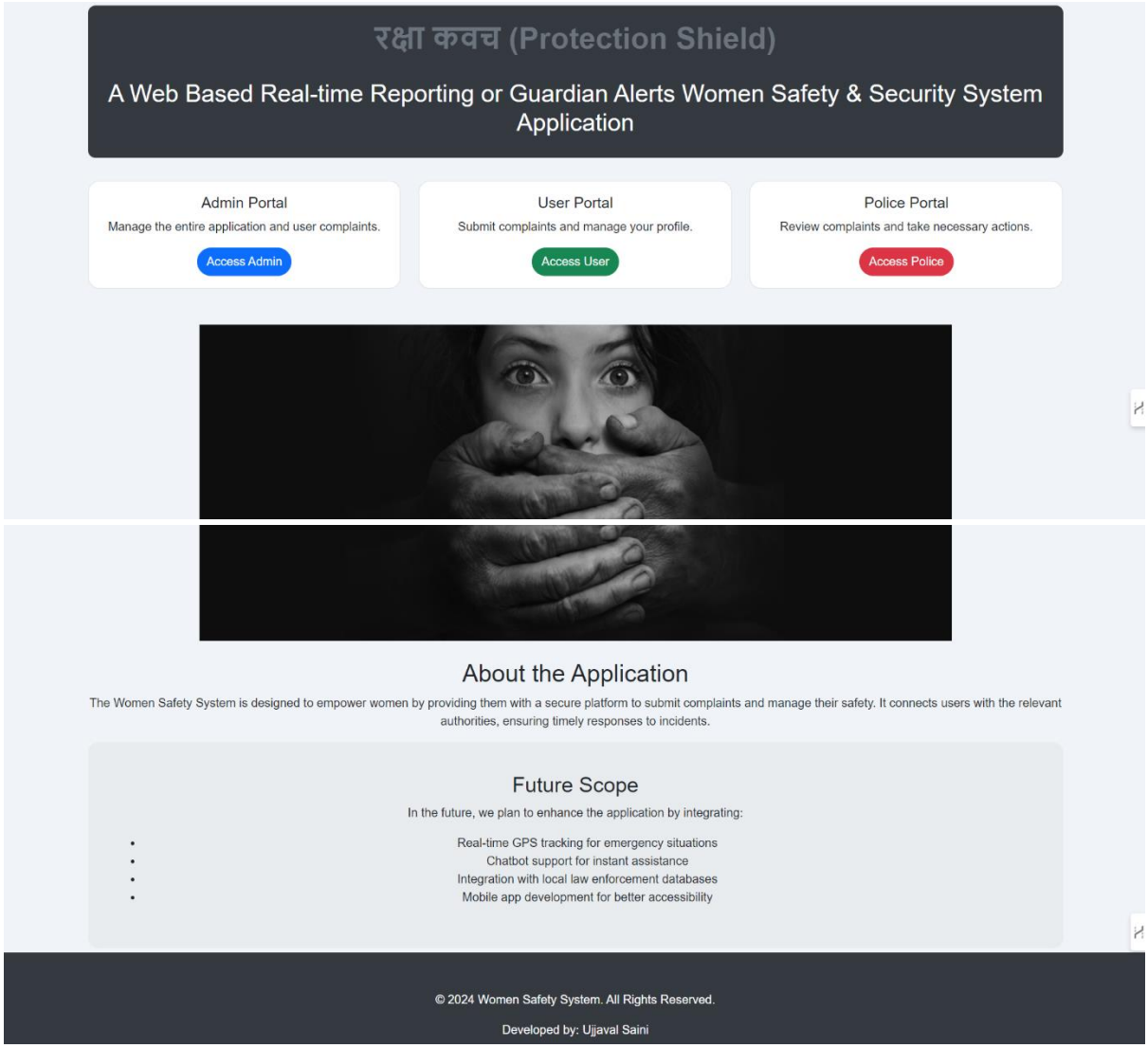
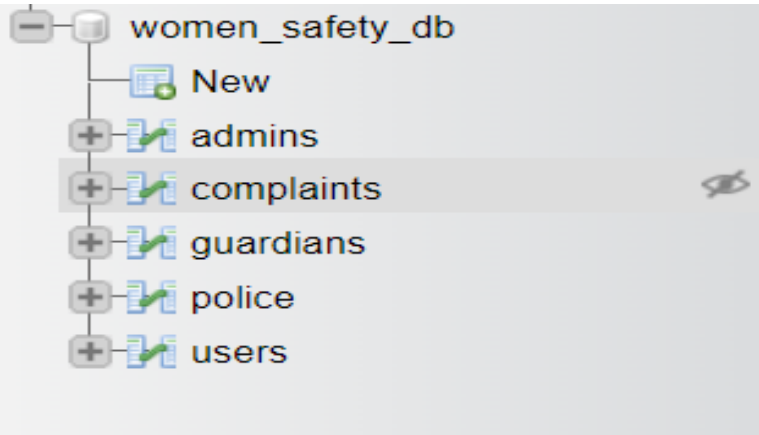


Figure: Schema Diagram

- Include a schema diagram that illustrates the database structure and relationships among tables such as users, complaints, alerts, etc.



-

Complaint:

	id	user_id	police_id	complaint_type	complaint_details	status	date_created
Edit Copy Delete	1	6	NULL	Harassment	help me	Pending	2024-10-25 03:51:04
Edit Copy Delete	2	6	NULL	Harassment	help me	Pending	2024-10-25 03:51:04
Edit Copy Delete	3	6	NULL	Others	kuch nhu hua h, its a test complaint	In Progress	2024-10-25 03:51:04
Edit Copy Delete	4	11	NULL	Robbery	someone stole my money	Pending	2024-10-25 06:41:32
Edit Copy Delete	5	11	NULL	Robbery	someone stolen my money	Pending	2024-10-25 11:13:46
Edit Copy Delete	6	11	NULL	Harassment	nothing, just testing	Pending	2024-10-25 11:53:52

Guardian:

	id	user_id	guardian_name	guardian_phone
Edit Copy Delete	1	6	saini	09690020293
Edit Copy Delete	2	11	govind kumar	09690020293
Edit Copy Delete	3	11	gorav	9690020293

Police:

	id	name	station	email	phone_number	password
Edit Copy Delete	3	suraj	sadar thana	gyananand90@gmail.com	9045229266	
Edit Copy Delete	4	sagar	pnjb	sagar@gmail.com	9757571218	
Edit Copy Delete	5	harsit	phg station	harsit@gmail.com	08470850200	
Edit Copy Delete	6	harsit	phg station	harsit@gmail.com	08470850200	$2y$10$ZonWdefpkytDU/2vxUKH3OZel/7ksDnGbuF/CkFSYEE...
Edit Copy Delete	7	shivam	up station	shivam@gmail.com	893434584732	$2y$10$DWjMi88x2E2c0CVm9qxlguYUM2D9uEhFGuSrHIHcpiF...

User:

				id	name	email	password	phone_number
Edit	Copy	Delete		1	Ujjaval Saini	sainiujvlpc@gmail.com	$2y$10$aOUZx1FSPwtk3x7z9Kf7TumAJtRKcb46pfMzwoHW0O3...	09690020293
Edit	Copy	Delete		2	aditya raj	adityagaurajraj@gmail.com	$2y$10$Fm3w9fbZGzIrERl7CSR2Fe2p2EVKz7QoCLRQPc.7YKh...	552005845124
Edit	Copy	Delete		6	radha	radha@gmail.com	$2y$10$DnBFk2VJ3peKTEETOZQ64e1WpPi6W46Zm35d8QioH9V...	12345
Edit	Copy	Delete		10	test1	test1@gmail.com	$2y$10$txnA2AR3QT68jJvn6LgrGORFatyoFBa.XXqVtXib42H...	
Edit	Copy	Delete		11	Aman galav	aman@gmail.com	$2y$10$YOlQdarCPdfL3el1/QgMautvPG1YXgWdmFafEHGk60M...	9012218389

The admin table does not contain any data because as a developer or admin, I am using a manual login to the admin portal.

Chapter 2: Tasks and Activities Performed

2.1 Profile of Problems

Initial research and user interviews revealed several key problems women face regarding safety, including:

- **Lack of Immediate Help:** Traditional methods of reaching out for help, such as calling a police station or guardian, often lead to delays that can be critical.
- **Fear of Reporting Incidents:** Many women are hesitant to report incidents due to societal stigma or fear of not being taken seriously.
- **Inadequate Communication Channels:** Existing systems for reporting harassment or violence are often not user-friendly or accessible, leading to underreporting of incidents.

2.2 Structure of the Project

The project comprises multiple modules:

1. User Module

This module is designed for the end-users (women) who need to utilize the safety features of the application.

- **User Registration & Login**
  - Functionality to register new users and authenticate existing users.
  - Password recovery options.
- **User Profile Management**

    - View and edit user profile information.
    - Option to manage guardian contacts.
- **Submit Complaint**
    - Form to report incidents (e.g., harassment, assault).
    - Fields for complaint type, details, and timestamp.
    - Validation for input data.
- **SOS Feature**
    - Button to send an SOS alert to registered guardians.
    - Sends SMS notifications with the user's location.
    - Triggers a notification in the system for further actions.
- **View Notifications**
    - Display notifications sent to the user about complaints, safety tips, or responses from authorities.
    - Filter options (e.g., read/unread).

2. Admin Module

This module is for administrators who manage user accounts and oversee complaint handling.

- **Admin Login**
    - Secure login for admin access.
- **User Management**
    - View, edit, and delete user accounts.
    - Access user complaint history.
- **Complaint Management**
    - Review and manage submitted complaints.
    - Update complaint status (e.g., resolved, in progress).

    - Option to send notifications to users regarding complaint updates.
- **View Notifications**
    - Access a dashboard of notifications sent to users.
    - Filter by status (e.g., sent, pending).

3. Police Module

This module allows police personnel to access and manage complaints related to incidents reported.

- **Police Login**
    - Secure login for police officers.
- **Manage Complaints**
    - View complaints reported by users.
    - Update complaint statuses based on investigation results.
    - Communicate with users regarding complaint progress.
- **View Notifications**
    - Access notifications about user incidents and alerts sent by the system.
    - Update the system on actions taken regarding complaints.

2.3 Scope and Feasibility

The project scope includes:

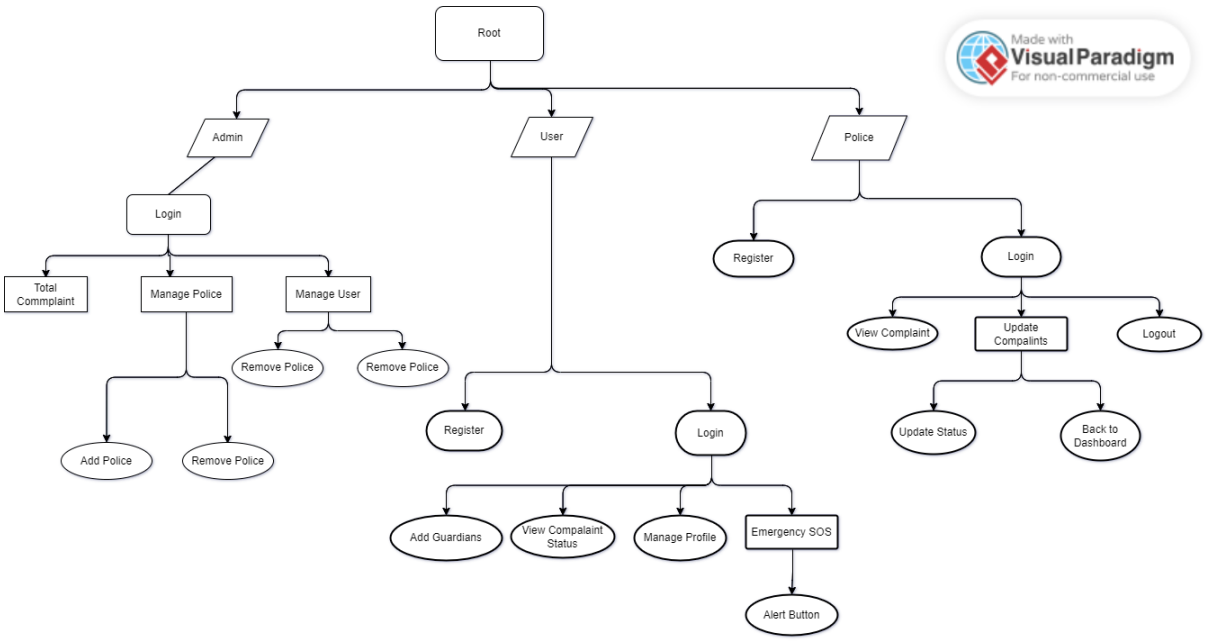
- **User Interaction:** Providing a user-friendly interface for women to report incidents and receive notifications.
- **Admin Oversight:** Allowing admins to monitor and manage incidents efficiently.
- **Police Engagement:** Enabling law enforcement to respond to complaints and follow up effectively.

The feasibility study concluded that the project is viable given the available resources, technology, and stakeholder support. Moreover, the development costs are justified by the potential impact on women's safety.

2.4 System Analysis

A thorough analysis of the system was conducted to identify user needs, technical requirements, and potential challenges. Key findings included:

- **User Roles:** Clearly defined user roles for women, admins, and police personnel.
- **Functional Requirements:** Key functionalities required for each role, including complaint submission, notification systems, and user management.
- **Non-functional Requirements:** Considerations for system performance, security, and usability were also critical.



2.5 System Design

The system design phase involved creating detailed designs and specifications:

- **User Interface (UI):** Designing wireframes for each module, focusing on ease of use and accessibility.

- **Database Design:** Establishing a relational database schema to manage user data, complaints, notifications, and police station information.
- **Architecture:** Develop an architecture that supports modular development and integrates with external APIs smoothly.

2.6 Implementation

The application was implemented using PHP, MySQL, and Bootstrap. Each module was developed and tested iteratively to ensure functionality and security.

2.7 Test Generation

Testing was a continuous process throughout development:

- **Unit Testing:** Each module was tested individually to ensure all functionalities were working as intended.
- **Integration Testing:** Ensuring that different modules worked together seamlessly, particularly the interaction between the application and the Twilio API.
- **User Acceptance Testing (UAT):** Feedback was gathered from potential users to identify any usability issues and necessary improvements before deployment.

Screen Shots

- (Include screenshots of various application interfaces, such as the user dashboard, complaint submission form, and notifications.)

रक्षा कवच (Protection Shield)

A Web Based Real-time Reporting or Guardian Alerts Women Safety & Security System Application

Admin Portal
Manage the entire application and user complaints.

Access Admin

User Portal
Submit complaints and manage your profile.

Access User

Police Portal
Review complaints and take necessary actions.

Access Police

About the Application

The Women Safety System is designed to empower women by providing them with a secure platform to submit complaints and manage their safety. It connects users with the relevant authorities, ensuring timely responses to incidents.

Future Scope

In the future, we plan to enhance the application by integrating:

- Real-time GPS tracking for emergency situations
- Chatbot support for instant assistance
- Integration with local law enforcement databases
- Mobile app development for better accessibility

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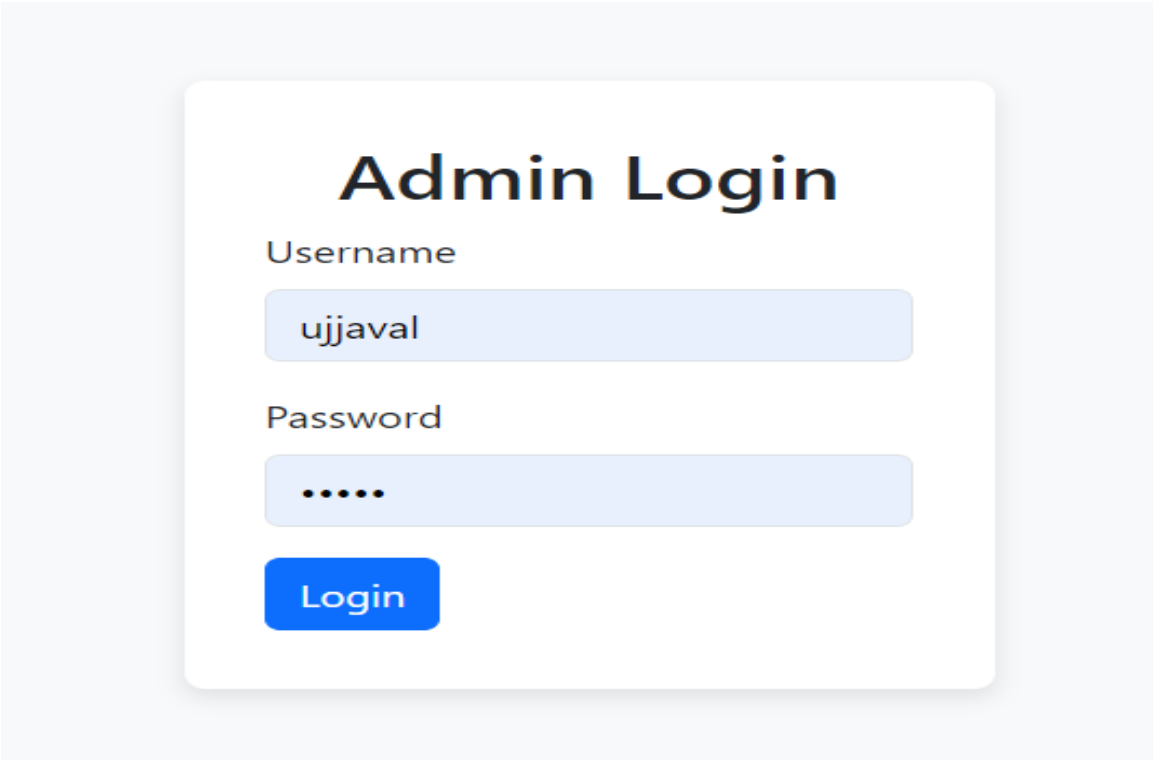
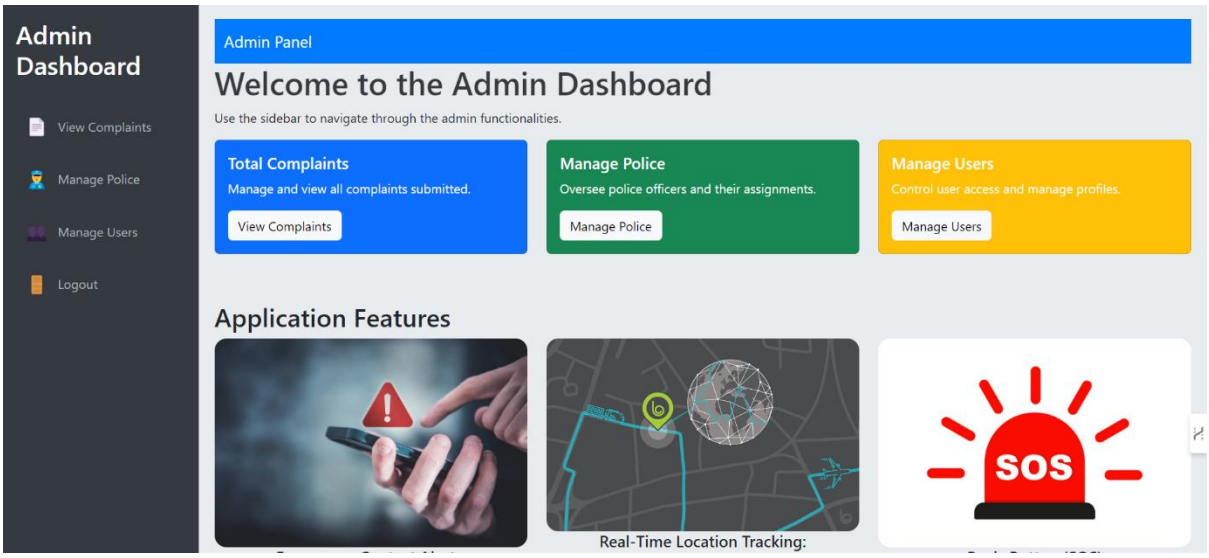
Developed by: Ujjaval Saini



ADMIN DASHBOARD



VIEW COMPLAINTS THROUGH ADMIN:

View Complaints

Dashboard Logout

Complaint Type	Details	Status	Date Submitted	User ID
Harassment	help me	Pending	2024-10-25 03:51:04	6
Harassment	help me	Pending	2024-10-25 03:51:04	6
Others	kuch nhu hua h, its a test complaint	In Progress	2024-10-25 03:51:04	6
Robbery	someone stole my money	Pending	2024-10-25 06:41:32	11
Robbery	someone stolen my money	Pending	2024-10-25 11:13:46	11
Harassment	nothing, just testing	Pending	2024-10-25 11:53:52	11

MANAGE POLICE THROUGH ADMIN:

Manage Police

Dashboard Logout

Officer Name

Station Name

Officer Email

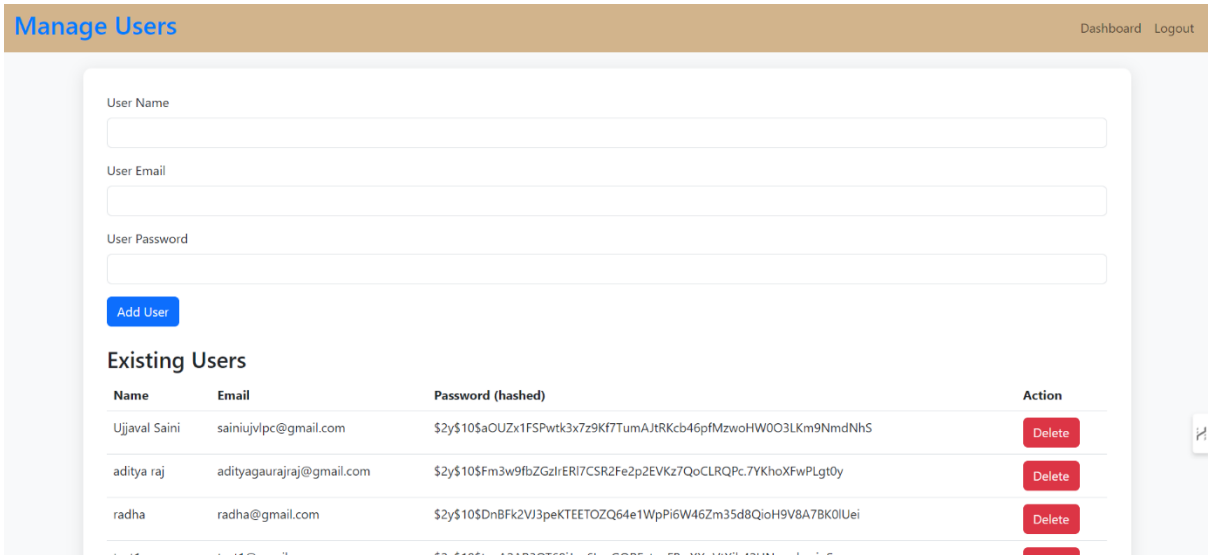
Officer Phone Number

Add Police Officer

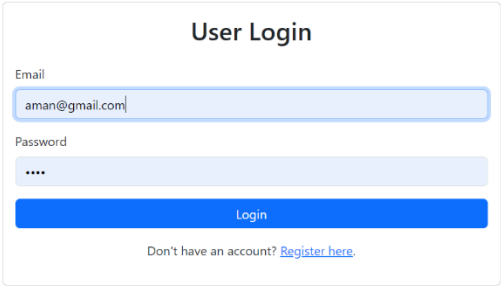
Existing Police Officers

Name	Station	Email	Phone Number	Action
suraj	sadar thana	gyananand90@gmail.com	9045229266	Delete
sagar	pnjb	sagar@gmail.com	9757571218	Delete

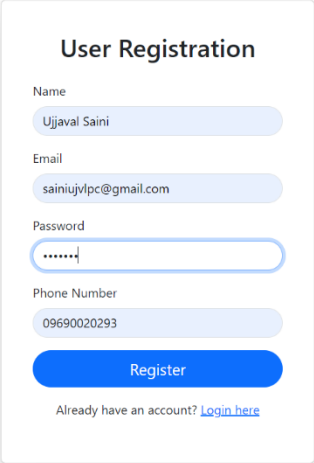
MANAGE USER THROUGH ADMIN:



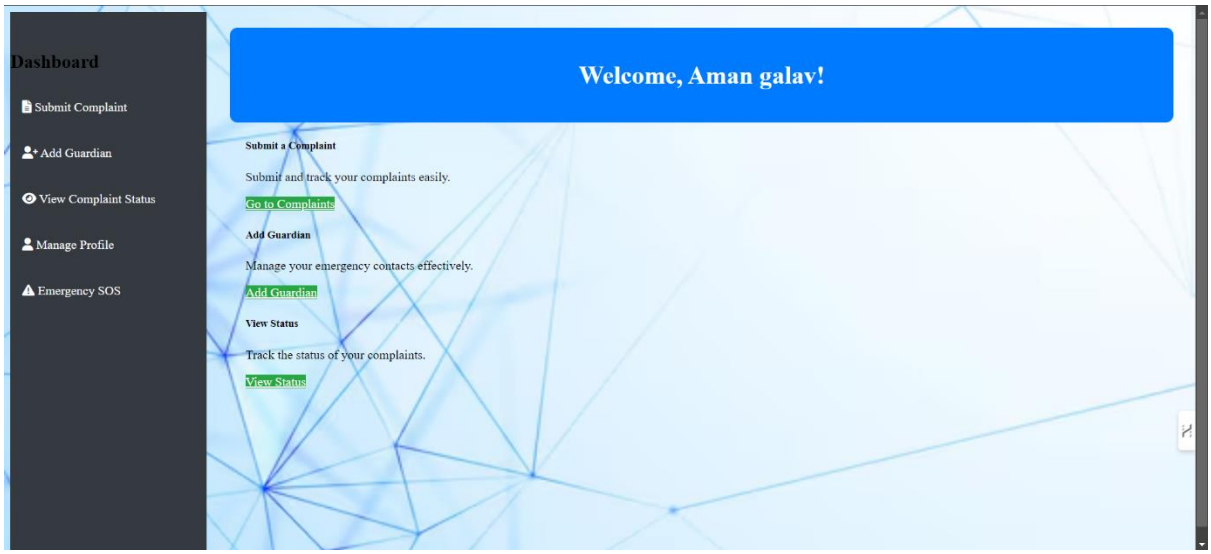
USER LOGIN:



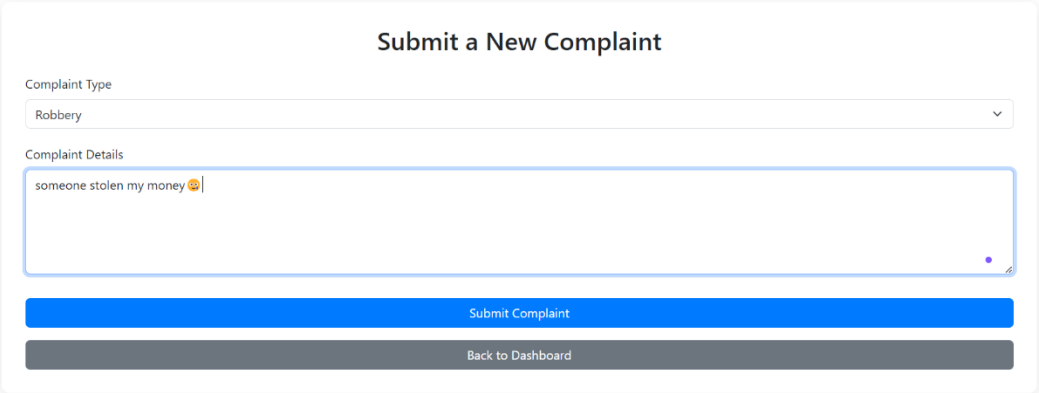
USER REGISTRATION:



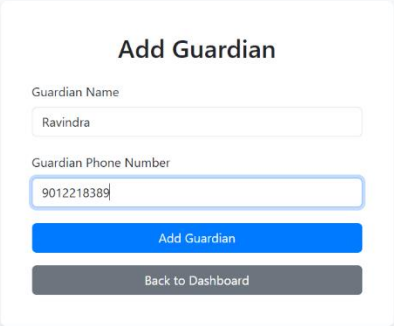
USER DASHBOARD:



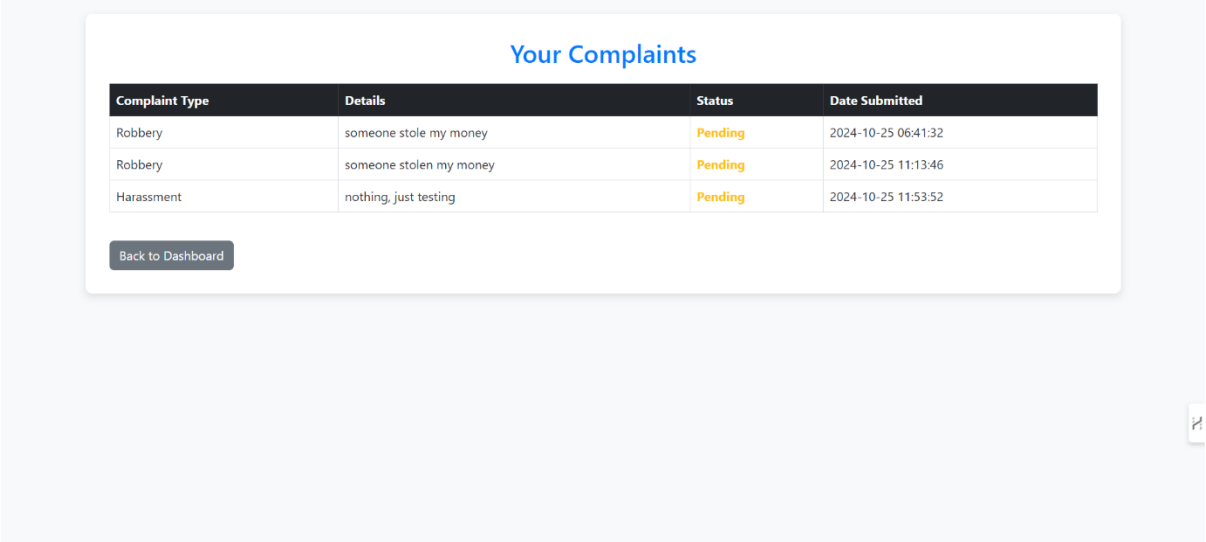
SUBMIT COMPLAINT THROUGH USER:



ADD GUARDIAN THROUGH USER:



VIEW YOUR COMPLAINT THROUGH USER:

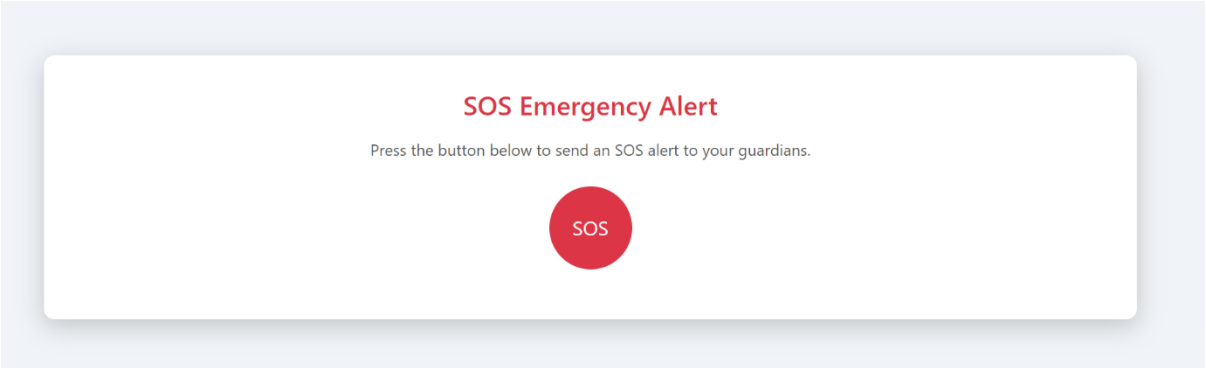


Your Complaints

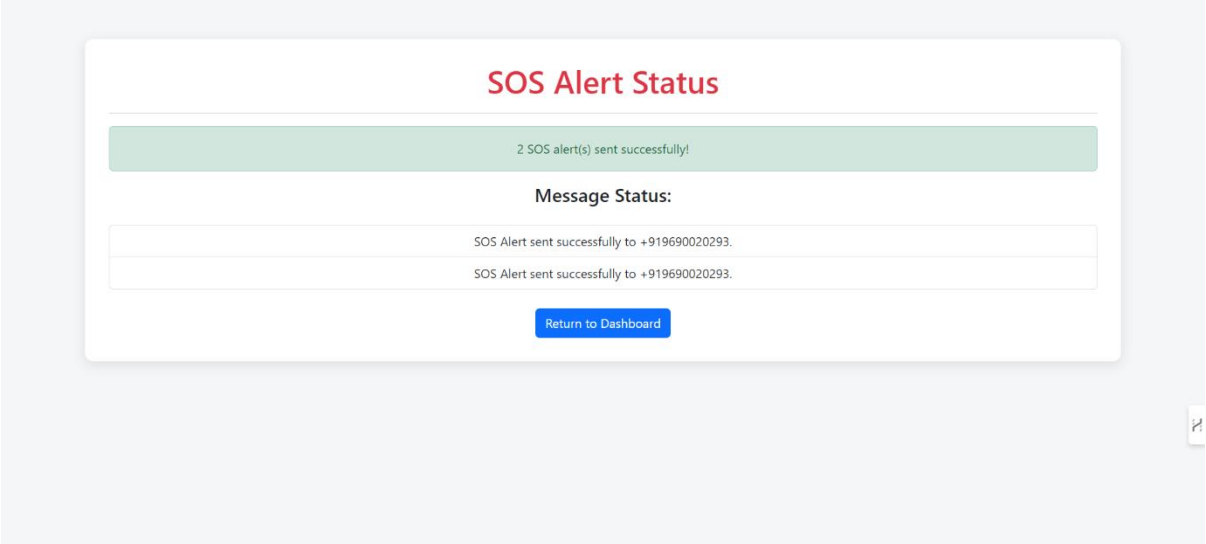
Complaint Type	Details	Status	Date Submitted
Robbery	someone stole my money	Pending	2024-10-25 06:41:32
Robbery	someone stolen my money	Pending	2024-10-25 11:13:46
Harassment	nothing, just testing	Pending	2024-10-25 11:53:52

Back to Dashboard

SOS MESSAGE THROUGH USER:



SOS STATUS:



SOS Alert Status

2 SOS alert(s) sent successfully!

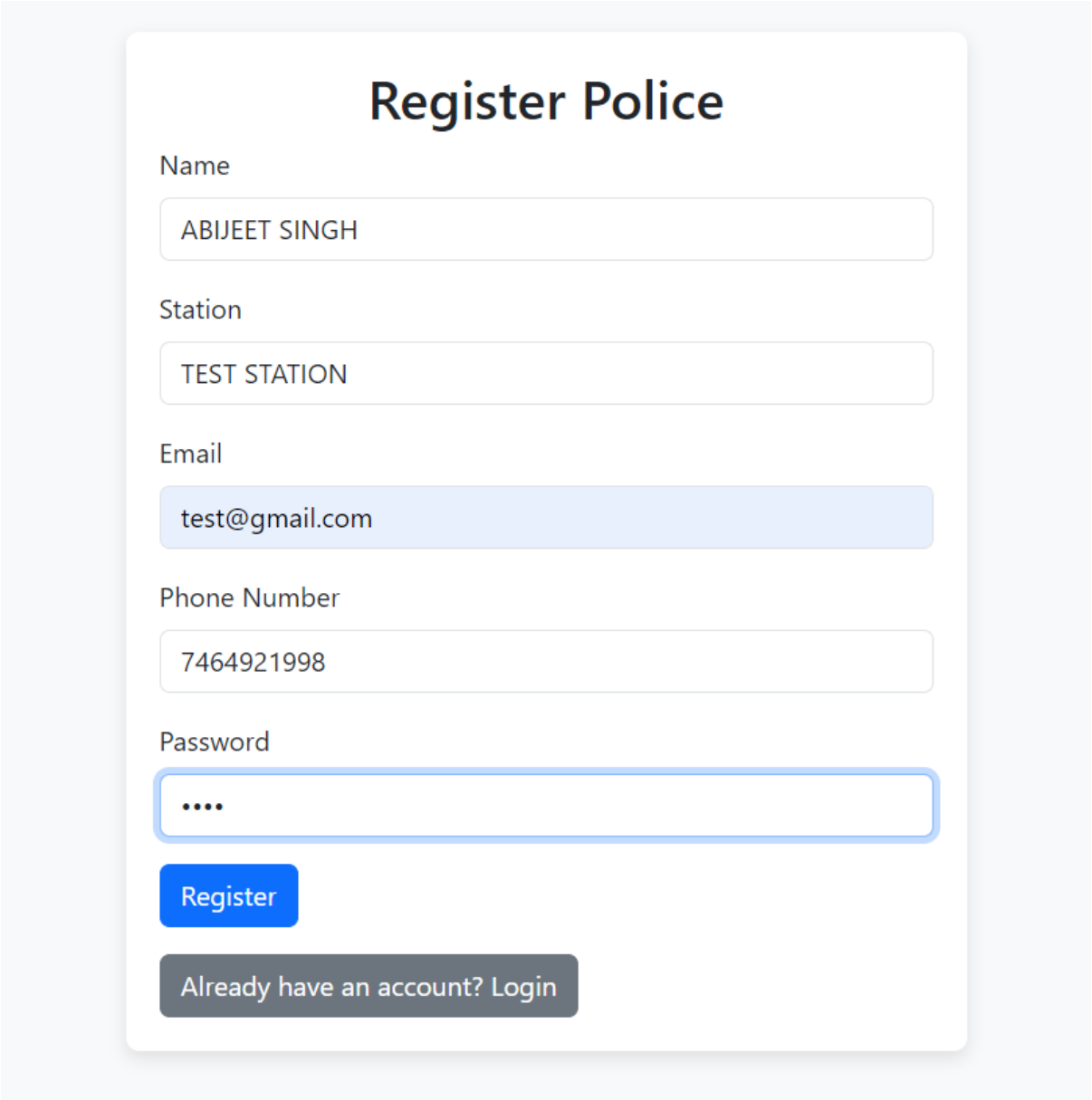
Message Status:

SOS Alert sent successfully to +919690020293.

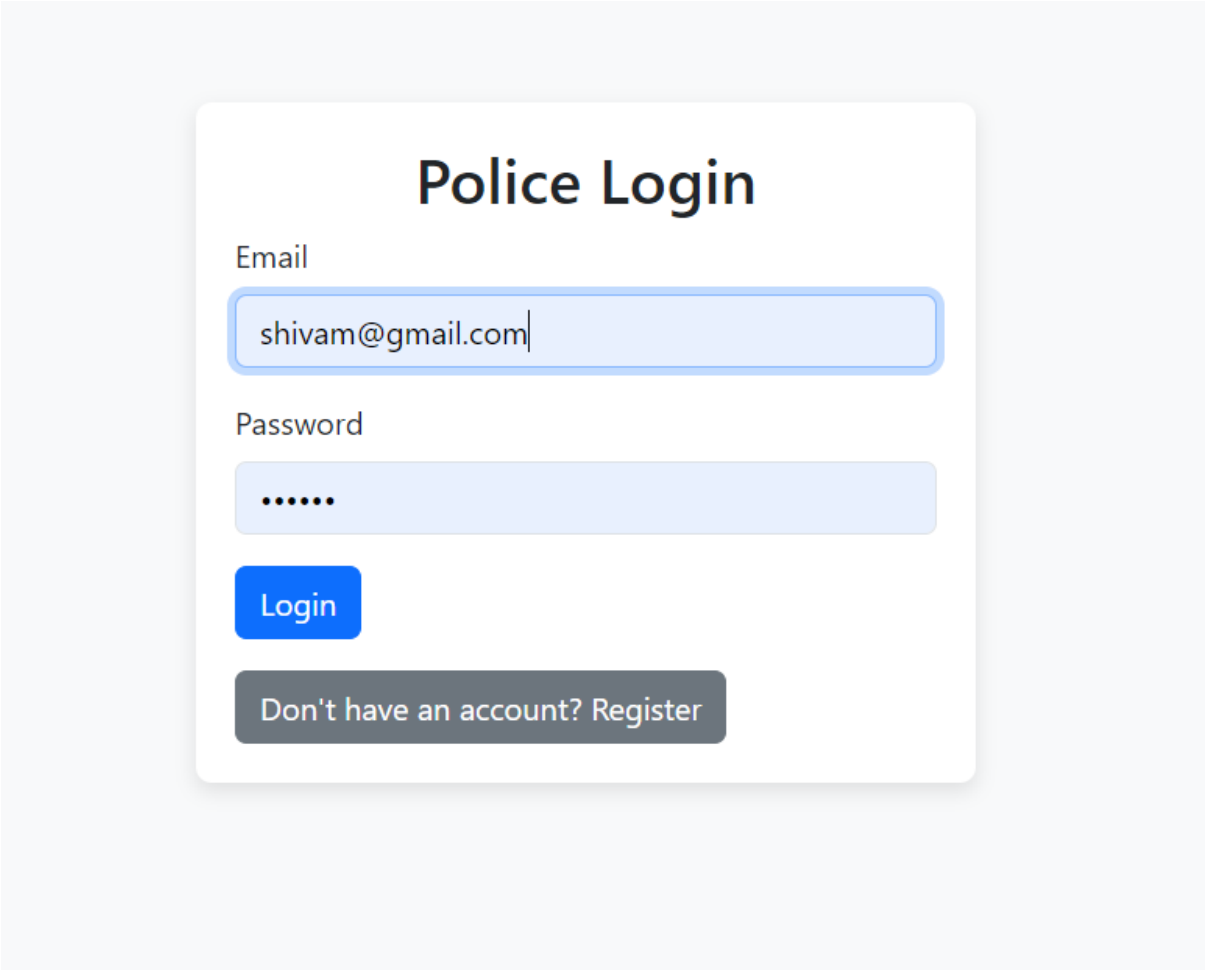
SOS Alert sent successfully to +919690020293.

Return to Dashboard

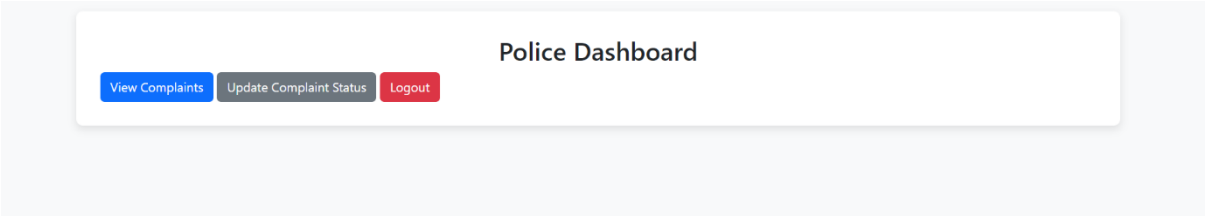
POLICE REGISTRATION:



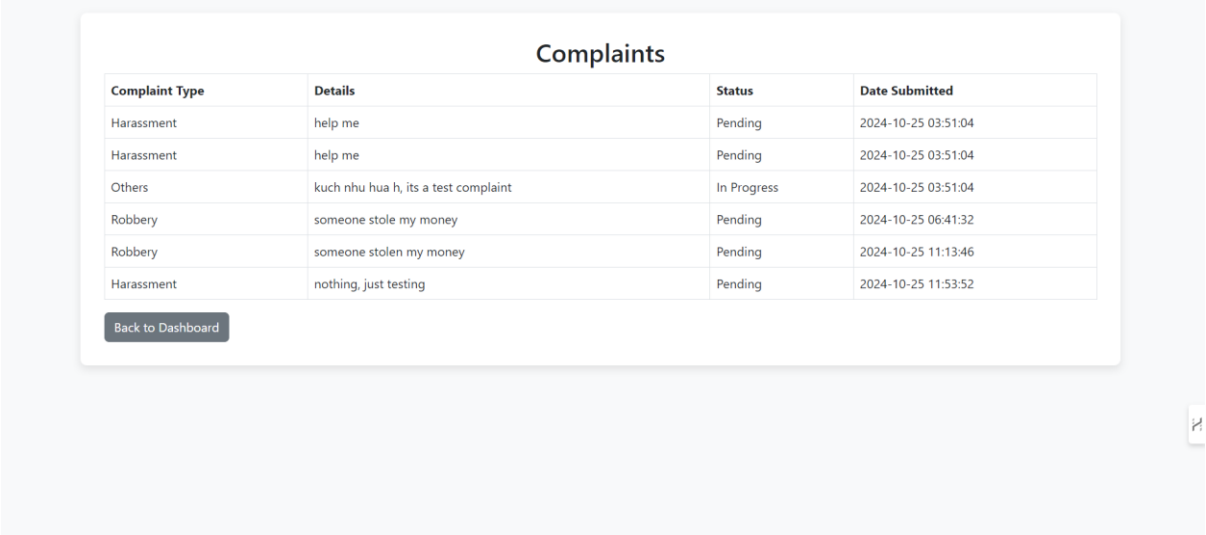
POLICE LOGIN:



POLICE DASHBOARD:



VIEW COMPLAINTS THROUGH POLICE:



Complaints

Complaint Type	Details	Status	Date Submitted
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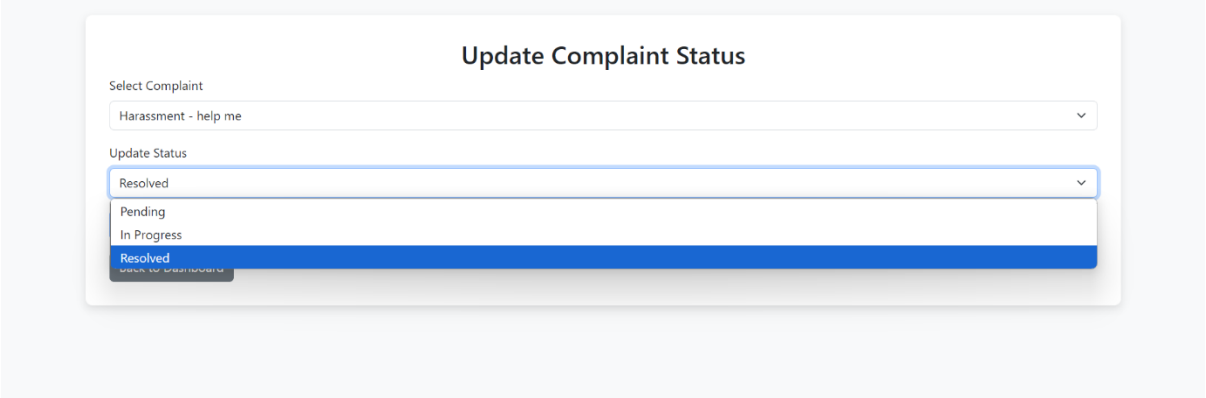
Robbery	someone stolen my money	Pending	2024-10-25 11:13:46
Harassment	nothing, just testing	Pending	2024-10-25 11:53:52

Back to Dashboard

UPDATE COMPLAINTS STATUS THROUGH POLICE:



2.8 Conclusion:

The Women Safety & Security System project aims to empower women by providing them with the tools necessary to report incidents and seek help effectively. By integrating user-friendly features with robust security measures, the application not only fosters a sense of security but also encourages proactive community engagement in ensuring women's safety. As society continues to address the critical issue of women's safety, this application serves as a vital resource in promoting awareness, support, and swift action in response to emergencies.

Future enhancements could include additional features such as real-time location tracking during an SOS alert, integration with local law enforcement databases for faster response times, and a feedback mechanism for users to report their experiences with the system. Overall, the Women Safety & Security System represents a significant step forward in utilizing technology to protect and support women in society.